

动理学方程的高效特征线型渐近保持方法

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摘 要: 动理学方程是描述微观粒子运动规律的重要数学模型, 其多尺度和高维度等特性为数值格式的设计带来显著挑战. 渐近保持格式是一类重要的多尺度计算框架, 旨在整个计算域内使用统一的数值求解器, 自动捕捉方程的渐近极限. 近年来, 我们以特征线回溯思想为基础, 发展了一类全尺度一致无条件稳定的渐近保持算法框架, 它具备无条件稳定、计算复杂度低、高度可并行和良好可扩展等优点, 特别适合高维跨尺度输运问题的长时间数值模拟, 将在国防工程、天体物理、半导体和生物化学等领域有广阔的应用前景.

关键词: 动理学方程; 并行计算; 特征线回溯法; 渐近保持格式

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0 引言

动理学方程是描述微观粒子运动规律的重要数学模型, 在辐射输运^[1]、等离子体物理^[2]、半导体^[3]和生物化学^[4]等领域具有广泛应用. 粒子在自由输运时不断发生碰撞, 其演化过程在统计意义下可由微分方程描述:

$$\frac{\partial f}{\partial t} + T(f) = C(f) .$$

其中, $f = f(t, x, v)$ 为粒子分布函数, $t \geq 0$ 为时间变量, $x \in \mathbb{X}$ 为位置变量, $v \in \mathbb{V}$ 为速度变量, $\mathbb{X} \times \mathbb{V}$ 为相空间, T 为输运算子, C 为碰撞算子.

动理学方程具有双曲-抛物混合特性. 传统数值离散多采用直线型方法 (method of lines): 先离散相空间, 再离散时间. 时间离散常采用隐式方法、预估校正法或特征线法, 以缓解刚性项带来的时间步长约束. 相空间离散可分为确定性方法与随机性方法. 在确定性方法中, 矩方法 (moment method) 和离散速度法 (discrete velocity method) 均是先对速度变量 v 采用全局近似, 再对位置变量 x 采用分片近似. 矩方法^[5,6]通过对速度变量进行矩展开, 利用正则化思想构造矩封闭条件^[7-11], 将高维动理学模型约简成低维宏观模型. 然而, 宏观模型的输运结构更加复杂, 在输运占优问题中易出现强波动性^[12,13]. 离散速度法^[14]在有限个离散速度点上直接求解动理学方程, 可严格保持输运结构. 但是, 由于信息传播方向受限于预设速度集, 当速度点数不足时易出现射线效应^[15,16]. 随机性方法的代表为蒙特卡罗法^[17] (Monte-Carlo method). 该方法将动理学方程等价表述为粒子随机运动过程, 利用随机采样构造数值解, 具有无网格性与高度并行性, 但在强碰撞区域计算效率会显著下降.

动理学方程具有多尺度特征. 为分析其长时间演化行为, 可通过量纲分析引入无量纲参数 Knudsen 数 ε , 并考虑下列多尺度模型:

$$\varepsilon \frac{\partial f}{\partial t} + T(f) = \frac{\varepsilon}{\lambda(\varepsilon)} C(f) .$$

当 $\lambda(\varepsilon) \sim \varepsilon$ 时, 输运效应与碰撞效应处于相同量级, 当 $t/\varepsilon \rightarrow +\infty$ 时, 方程收敛至稳态极限. 当 $\lambda(\varepsilon) \sim \varepsilon^2$ 时, 碰撞效应显著增强但仍不足以完全抹平输运效应, 方程呈现双曲-抛物混合特性: 当 $\varepsilon \gg 0$ 时, 方程是双曲占优的; 当 $\varepsilon \approx 0$ 时, 方程是抛物占优的; 当 $t \rightarrow +\infty$ 时, 方程仍然收敛至稳态极限. 为构造能够同时适用于不同区域的

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数值方法, 现有研究主要沿两个方向发展: 区域分解(domain decomposition)与渐近保持(asymptotic-preserving). 区域分解方法^[18-24]将计算域划分为不同的物理子区域, 在各区域分别采用适合的数值方法求解, 并通过界面条件实现整体匹配. 渐近保持方法^[25]旨在整个计算域内采用统一的数值方法, 在离散意义下自动捕捉方程的渐近极限. 典型的渐近保持格式有: 1) 松弛格式^[26,27]; 2) 宏微观分解^[28-34]; 3) 奇偶分解^[35,36]; 4) 罚函数法^[37,38]; 5) 指数积分法^[39-43]等等.

近年来, 我们以线性动理学方程为出发点, 围绕特征线回溯法开展了一系列研究, 发展了一类全尺度无条件稳定、高度可并行且易扩展的渐近保持算法, 并进行了相应的理论稳定性分析. 同时, 我们进一步将该算法推广应用于非线性辐射输运方程, 取得了良好的数值效果.

1 基于特征线回溯法的渐近保持格式

对于具有抛物极限的多尺度动理学模型, 其输运项和碰撞项在 $\varepsilon \approx 0$ 时均具有强刚性, 这为时间离散带来了显著的数值挑战. 一方面, 根据 CFL 条件, 显式格式受到非常严苛的时间步长约束 $\tau \sim \varepsilon h$, 其中 τ 为时间步长, h 为空间尺寸; 另一方面, 隐式格式虽然无条件稳定, 但在高维情形下需求解大规模强耦合线性系统, 计算成本极其高昂. 为平衡稳定性和计算成本, Peng 等^[32-34]构造了一类显隐格式, 通过隐式处理扩散相关项, 实现抛物占优区域 ($\varepsilon \ll h$) 无条件稳定. 但是, 在过渡区域 ($\varepsilon \sim h$) 仍然会受限于抛物型时间步长约束 ($\tau \sim h^2$) 对跨区域问题而言仍存在明显局限. 注意到这一瓶颈的根源在于输运项的刚性, 我们发展了一类基于特征线回溯法的渐近保持格式^[44,45], 其具备如下优势:

1) 全尺度一致无条件稳定性且计算复杂度低. 与显式格式相比, 该方法在理论上不受 CFL 步长约束; 与隐式格式相比, 仅需求解少量小规模宏观线性系统, 计算代价显著降低; 与显隐格式相比, 在过渡区域仍可保持无条件稳定, 从而显著提升跨区域问题的计算效率;

2) 高度并行性. 在传统隐式迎风格式中, 由于目标时间层单元在位置变量上具有强耦合性, 导致算法关于位置变量的并行效率低下. 而特征线回溯法可完全解耦目标时间层单元, 使算法在整个相空间上具有近似线性的加速比;

3) 良好的可扩展性与兼容性. 该方法对具有刚性输运项的动理学模型均具有普适性. 此外, 我们已经将其应用至蒙特卡罗法^[46, 47]和矩方法^[48]框架, 显著提升了相应方法在长时间模拟时的稳定性和计算效率.

1.1 模型重构

下面我们以一维方程为例介绍基于特征线法的模型重构过程. 设 $E(v)$ 是全局平衡态函数, 满足归一化条件和对称性条件:

$$\int_{\mathbb{V}} E(v) dv = 1, \quad \int_{\mathbb{V}} v E(v) dv = 0. \quad (1)$$

设 $f_{eq}(t, x, v) = \rho(t, x) E(v)$, $\rho(t, x) = \int_{\mathbb{V}} f(t, x, v) dv$, 考虑一维线性输运方程:

$$\varepsilon \frac{\partial f}{\partial t}(t, x, v) + v \frac{\partial f}{\partial x}(t, x, v) = \frac{\varepsilon \sigma}{\lambda(\varepsilon)} (f_{eq}(t, x, v) - f(t, x, v)). \quad (2)$$

设 $j(t, x) = \int_{\mathbb{V}} v f(t, x, v) dv$, 系统满足局部质量守恒律:

$$\varepsilon \frac{\partial \rho}{\partial t}(t, x) + \frac{\partial j}{\partial x}(t, x) = 0. \quad (3)$$

设 $f_{\perp}(t, x, v) = f(t, x, v) - f_{eq}(t, x, v)$, 由 f_{eq} 的定义和 E 的归一化条件得

$$\int_{\mathbb{V}} f_{\perp}(t, x, v) dv = 0. \quad (4)$$

引入参数 τ , 由特征线法得

$$f(t, x, v) = e^{-\sigma \tau / \lambda} f(t - \tau, x - \frac{v}{\varepsilon} \tau, v) + \int_0^{\tau} \frac{\sigma}{\lambda} e^{-\sigma(\tau-s)/\lambda} f_{eq}(t - \tau + s, x - \frac{v}{\varepsilon}(\tau - s), v) ds. \quad (5)$$

引入参数 $\theta = \theta(\sigma, \lambda) \in [0, 1]$ 满足: 1) 当 $\sigma \rightarrow 0$ 时, $\theta \rightarrow 1$; 2) 当 $\lambda \sim \varepsilon$ 时, $\theta \approx 1$; 3) 当 $\lambda \sim \varepsilon^2$ 或 $\sigma \rightarrow +\infty$ 时, $\theta \approx 0$. 由 (5) 可将 $f_{\perp}(t, x, v)$ 分解为凸组合

$$f_{\perp}(t, x, v) = \theta f_{\perp, H}(t, x, v) + (1 - \theta) f_{\perp, P}(t, x, v) . \quad (6)$$

一方面, $f_{\perp, H}(t, x, v)$ 用于捕捉自由输运极限和稳态极限, 写作

$$f_{\perp, H}(t, x, v) = e^{-\sigma\tau/\lambda} (f_{\perp} + f_{eq})(t - \tau, x - \frac{v}{\varepsilon}\tau, v) - e^{-\sigma\tau/\lambda} f_{eq}(t, x, v) + r_H(t, x, v) . \quad (7)$$

定义

$$\varphi_k(z) = \begin{cases} 1, k = 0; \\ \int_0^1 \frac{\omega^{k-1}}{(k-1)!} e^{\omega z} d\omega, k \geq 1. \end{cases} \quad (8)$$

其满足

$$\varphi_k(0) = \frac{1}{k!}, \quad \forall k \geq 0 , \quad (9)$$

且

$$\varphi_k(z) = \frac{e^z \varphi_{k-1}(0) - \varphi_{k-1}(z)}{z}, \quad \forall k \geq 1 . \quad (10)$$

记 $\phi_k(z) = z^k \varphi_k(z)$, $D^k = \left(\frac{\lambda}{\sigma}\right)^k \left(\frac{\partial}{\partial t} + \frac{v}{\varepsilon} \frac{\partial}{\partial x}\right)^k$, 则残差项 $r_H(t, x, v)$ 可写作

$$r_H(t, x, v) = -e^{-\sigma\tau/\lambda} \sum_{k \geq 1} \phi_{k+1}(-\frac{\sigma\tau}{\lambda}) D^k f_{eq}(t, x, v) . \quad (11)$$

另一方面, $f_{\perp, P}(t, x, v)$ 由分部积分法得到, 用于捕捉抛物极限, 写作

$$f_{\perp, P}(t, x, v) = e^{-\sigma\tau/\lambda} f_{\perp}(t - \tau, x - \frac{v}{\varepsilon}\tau, v) - \frac{\lambda}{\sigma} (1 - e^{-\sigma\tau/\lambda}) \left(\frac{\partial f_{eq}}{\partial t} + \frac{v}{\varepsilon} \frac{\partial f_{eq}}{\partial x}\right)(t, x, v) - r_P(t, x, v) , \quad (12)$$

其中, 残差项 $r_P(t, x, v)$ 可写作

$$r_P(t, x, v) = -\frac{\lambda}{\sigma} e^{-\sigma\tau/\lambda} \sum_{k \geq 1} \phi_{k+1}(-\frac{\sigma\tau}{\lambda}) D^{k+1} f_{eq}(t, x, v) . \quad (13)$$

将上述特征线解代入 (3), 联立 (2) 可得

$$\begin{cases} \frac{\partial f}{\partial t}(t, x, v) + \frac{v}{\varepsilon} \frac{\partial f}{\partial x}(t, x, v) = \frac{\sigma}{\lambda} (f_{eq}(t, x, v) - f(t, x, v)), \\ \frac{\partial \rho}{\partial t}(t, x) + \frac{1}{\varepsilon} e^{-\sigma\tau/\lambda} \int_{\mathbb{V}} v \frac{\partial(f_{\perp} + \theta f_{eq})}{\partial x}(t - \tau, x - \frac{v}{\varepsilon}\tau, v) dv = \frac{\lambda}{\varepsilon^2 \sigma} (1 - \theta) (1 - e^{-\sigma\tau/\lambda}) \Theta \frac{\partial^2 \rho}{\partial x^2}(t, x) + R(t, x). \end{cases} \quad (14)$$

其中, $\Theta = \int_{\mathbb{V}} v^2 E(v) dv$ 为扩散系数, $R(t, x)$ 为残差项, 写作

$$R(t, x) = \int_{\mathbb{V}} v (\theta r_H(t, x, v) - (1 - \theta) r_P(t, x, v)) dv . \quad (15)$$

当 $\lambda(\varepsilon) = \varepsilon$ 时, 有 $\theta = 1$,

$$f(t, x, v) \xrightarrow{t/\varepsilon \rightarrow +\infty} f^{\infty}(v) = \rho^{\infty}(x) E(v) . \quad (16)$$

此时, ρ^{∞} 是由初值条件和边界条件确定的常数. 当 $\lambda(\varepsilon) = \lambda_0 \varepsilon^2$ 时, 有 $\theta \approx 0$,

$$f(t, x, v) \xrightarrow{\varepsilon \rightarrow 0} f^{\lambda_0}(t, x, v) = \rho^{\lambda_0}(t, x) E(v) , \quad (17)$$

其中, ρ^{λ_0} 满足抛物方程

$$\frac{\partial \rho^{\lambda_0}}{\partial t}(t, x) = \frac{\lambda_0}{\sigma} \Theta \frac{\partial^2 \rho^{\lambda_0}}{\partial x^2}(t, x) . \quad (18)$$

再令 $t \rightarrow +\infty$ 或 $\lambda_0 \rightarrow +\infty$, 可得 $\rho^{\lambda_0}(t, x) \rightarrow \rho^{\infty}(x)$. 对于更复杂的碰撞算子, 一种可行的策略是采用线性加罚对模型进行重构. 以各向异性散射为例, 其碰撞算子可写为

$$C(f) = \int_{\mathbb{V}} \sigma(v, v') (f(t, x, v') - f(t, x, v)) dv', \quad \int_{\mathbb{V}} \sigma(v, v') dv' = 1 . \quad (19)$$

可将其分解为

$$C(f) = P(f) + Q(f), P(f) = C(f) - Q(f), \quad (20)$$

其中, $P(f) = \sigma_0(f_{eq}(t, x, v) - f(t, x, v))$ 为罚项, $Q(f)$ 为源项. 应用前文所述的模型重构过程, 便能构造出渐近保持的宏观-微观输运模型.

1.2 数值格式

下面我们以一阶有限体积格式为例描述数值格式的构造过程. 设 τ_n 为第 n 个时间步的步长, $\tau = \max_n \tau_n$, x_i 是单元 $I_i = [x_{i-1/2}, x_{i+1/2}]$ 的中点, $h_i = x_{i+1/2} - x_{i-1/2}$, $h = \max_i h_i$, (v_k, w_k) 是第 k 个离散速度点与其对应的积分权重, ρ_i^n 是变量 ρ 在 t_n 时刻单元 I_i 上的均值, $f_{i,k}^n$ 是变量 f 在 t_n 时刻单元 I_i 上第 k 个离散速度分量的均值, $T_k^n x = x - \frac{v_k}{\varepsilon} \tau_n$ 是位置点 x 沿第 k 个离散速度回溯至 t_n 时刻的上游位置点. 数值格式的构造分为两个步骤: 宏观方程的离散和微观方程的离散. 整体算法流程可见图 1.

1) 宏观方程的离散. 时间上采用向后欧拉时间离散, 可得:

$$\frac{\rho_i^{n+1/2} - \rho_i^n}{\tau_n} + \frac{1}{\varepsilon} e^{-\sigma \tau_n / \lambda} \sum_k w_k v_k S_{h,k}^n (f_{\perp,i,k}^n + \theta f_{eq,i,k}^n) = \frac{\lambda}{\varepsilon^2 \sigma} (1 - \theta) (1 - e^{-\sigma \tau_n / \lambda}) \Theta D_h \rho_i^{n+1/2}. \quad (21)$$

这里可以看到, 我们对扩散项 $D_h \rho$ 采用隐式时间离散, 因而在抛物极限处能够克服抛物型时间步长约

束 $\tau \sim h^2$. 在宏观数值格式的构造过程中, 离散输运算子 $S_{h,k}^n$ 的构造是关键, 下面我们介绍两种典型思想:

a) 短程回溯. 该思想源自统一气体动力学格式^[39,43], 通过沿局部特征线回溯重构界面通量. 记 $g = f_{\perp} + \theta f_{eq}$, 定义

$$S_{h,k}^n g_{i,k}^n = \begin{cases} \frac{1}{h_i} (g_{i+1/2,k}^{n,l} - g_{i-1/2,k}^{n,l}), & v_k > 0; \\ \frac{1}{h_i} (g_{i+1/2,k}^{n,r} - g_{i-1/2,k}^{n,r}), & v_k < 0. \end{cases} \quad (22)$$

其中, $g_{i+1/2,k}^{n,l}$ 是通量左侧重构值, $g_{i+1/2,k}^{n,r}$ 是通量右侧重构值. 对于一阶格式而言, 可取 $g_{i+1/2,k}^{n,l} = g_{i,k}^n$ 和 $g_{i+1/2,k}^{n,r} = g_{i+1,k}^n$. 由于短程回溯时的特征线截断造成数值依赖域缩小, 导致该方法在过渡区域的稳定性条件为 $\tau \sim \varepsilon h$.

b) 长程回溯^[44,45]. 该思想源自半拉格朗日 (Semi-Lagrangian) 格式^[49,50], 通过严格追踪特征线至上游时间层, 利用上游单元重构追踪点处的信息. 因为数值依赖域严格包含微分方程依赖域, 所以该方法的稳定性不受时间步长约束. 假设 $T_k^n x \in I_m$, $g_{*,k}^n(x)$ 是基于 $\{g_{i,k}^n\}$ 信息重构多项式, 满足对于所有 $\ell \in \{0, 1\}$,

$$\begin{cases} \frac{1}{h_{m-\ell}} \int_{I_{m-\ell}} g_{*,k}^n(x) dx = g_{m-\ell,k}^n, & v_k > 0; \\ \frac{1}{h_{m+\ell}} \int_{I_{m+\ell}} g_{*,k}^n(x) dx = g_{m+\ell,k}^n, & v_k < 0. \end{cases} \quad (23)$$

由此可定义如下离散输运算子:

$$S_{h,k}^n g_{i,k}^n = \begin{cases} \frac{1}{h_i} (g_{*,k}^n(T_k^n(x_{i+1/2}^-)) - g_{*,k}^n(T_k^n(x_{i-1/2}^-))), & v_k > 0; \\ \frac{1}{h_i} (g_{*,k}^n(T_k^n(x_{i+1/2}^+)) - g_{*,k}^n(T_k^n(x_{i-1/2}^+))), & v_k < 0. \end{cases} \quad (24)$$

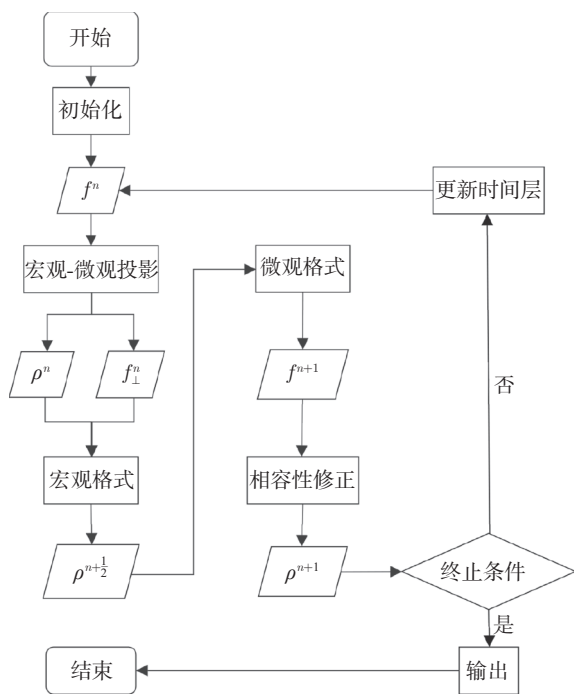


图 1 算法流程图.

Fig. 1 Algorithm flowchart.

2) 微观方程的离散. 对于无条件稳定的微观数值格式的构造过程, 我们介绍两种研究思路:

a) 全隐格式. 时间上采用向后欧拉时间离散, 可得:

$$\frac{f_{i,k}^{n+1} - f_{i,k}^n}{\tau_n} + \frac{v_k}{\varepsilon} B_{h,k} f_{i,k}^{n+1} = \frac{\sigma}{\lambda} (f_{eq,i,k}^{n+1} - f_{i,k}^{n+1}) . \quad (25)$$

其中, $B_{h,k}$ 是迎风离散算子, 其一阶形式如下:

$$B_{h,k} f_{i,k}^n = \begin{cases} \frac{1}{h_i} (f_{i,k}^n - f_{i-1,k}^n), & v_k > 0; \\ \frac{1}{h_i} (f_{i+1,k}^n - f_{i,k}^n), & v_k < 0. \end{cases} \quad (26)$$

对于输运方程离散后得到的线性代数系统, 常采用源迭代 (source iteration) 方法求解. 该方法的核心思想是预估 f_{eq}^{n+1} , 解耦各离散速度点对应的离散方程, 从而实现逐速度的并行求解. 为保证迭代法的收敛性, 为确保源迭代的收敛性, 需要构造合适的预条件子, 最典型的策略是合成加速型方法^[51-54] (synthetic acceleration). 为进一步提高迭代效率, 一方面可利用历史信息构造更优的搜索方向, 例如 Krylov 子空间方法^[55,56]; 另一方面也可借助多重网格方法^[57,58]在不同网格层次间传递误差以加速收敛. 此外, 若能获得高质量的初始迭代解, 也可显著减少迭代次数. 在我们的算法框架中, 宏观方程提供了高质量的数值局部平衡态 $f_{eq}^{n+1/2}$, 可视为特殊的初始迭代解, 仅需单步更新即可将残差降至目标精度. 关于输运方程迭代法的系统介绍, 可参考综述文章^[59].

b) 半拉格朗日格式. 时间上采用向后欧拉格式,

$$f_{i,k}^{n+1} = \frac{\lambda}{\lambda + \sigma\tau_n} S_{i,k}^n f_{i,k}^n + \frac{\sigma\tau_n}{\lambda + \sigma\tau_n} f_{eq,i,k}^{n+1} . \quad (27)$$

记 $f_{*,k}^n(x)$ 是基于 $\{f_{i,k}^n\}$ 信息的重构多项式, 由此可定义离散输运算子:

$$S_{h,k}^n f_{i,k}^n = f_{*,k}^n(T_k^n(x_i)) . \quad (28)$$

当 f_{eq}^{n+1} 给定时, 微观量的更新无需求解线性方程组, 且 t_{n+1} 时刻各空间单元之间不存在依赖关系, 从而能够实现逐单元、逐速度的完全并行求解, 特别适合在大规模并行集群上执行^[60,61]. 若令 $f_{eq}^{n+1} = f_{eq}^{n+1/2}$, 仅需单步更新即可得到目标精度的微观数值解. 关于高精度半拉格朗日格式的构造, 可参考文献^[62-64].

c) 蒙特卡罗格式. 假定数值解有如下表示:

$$f(t, x, v) = \sum_p \omega_p(t) \delta(x - x_p(t)) \delta(v - v_p(t)) , \quad (29)$$

其中, $x_p(t)$, $v_p(t)$ 和 $\omega_p(t)$ 分别是粒子 p 在 t 时刻的位置, 速度和权重, 它们满足以下关系:

$$\frac{dx_p}{dt} = \frac{v_p}{\varepsilon}, \quad \frac{dv_p}{dt} = 0, \quad \frac{d\omega_p}{dt} = -\frac{\sigma}{\lambda} \omega_p . \quad (30)$$

粒子的初始位置, 速度和时间通过随机抽样确定, 而其初始权重则依据给定的初值条件、边值条件及预估 f_{eq}^{n+1} 进行分配. 与需要根据指数分布解析处理粒子散射的传统蒙特卡罗方法相比, 本方法无需显式模拟每次散射, 而是将散射项拆分为阻尼项和源项. 这一改进在方差略有增加的情况下, 显著改善了计算效率, 达成了良好的性能权衡.

需要指出的是, 对于宏观方程 (21) 中的对流项计算, 我们可采用蒙特卡罗方法进行求解. 记 $g = f_{\perp} + \theta f_{eq}$, 可定义离散输运算子如下:

$$\bar{S}_h^n g_i^n = \frac{1}{\tau_n h_i} \left(\sum_{p \in A_{i+1/2}^n} \text{sign}(v_p) \omega_p - \sum_{p \in A_{i-1/2}^n} \text{sign}(v_p) \omega_p \right) , \quad (31)$$

其中, $A_{i+1/2}^n$ 表示在时间区间 $[t_n, t_{n+1}]$ 内穿过 $x_{i+1/2}$ 的所有粒子, 包括来自上一层的源粒子, 边界条件粒子以及由 f_{eq} 在 t_n 时刻放出的粒子. 从而, 宏观离散格式可改写为:

$$\frac{\rho_i^{n+1/2} - \rho_i^n}{\tau_n} + \frac{1}{\varepsilon} \bar{S}_h^n g_i^n = \frac{\lambda}{\varepsilon^2 \sigma} (1 - \theta) (1 - e^{-\sigma\tau_n/\lambda}) \Theta D_h \rho_i^{n+1/2} . \quad (32)$$

关于蒙特卡罗方法的实现细节, 可参考文献^[65,66].

3) 宏观-微观相容性. 若直接令 $\rho^{n+1} = \rho^{n+1/2}$, 将会造成数值模拟的不稳定, 其根源在于宏观量与微观量之间缺乏相容性, 即, $\rho^{n+1/2} \neq \sum_k w_k f_k^{n+1}$. 为此, 我们需要利用微观量对宏观量进行相容重构:

$$\rho_i^{n+1} = \sum_k w_k f_{i,k}^{n+1}. \quad (33)$$

2 在辐射输运问题中的应用

2.1 线性输运方程

下面我们给出线性输运方程的部分三维数值算例. 间断有限元方法是求解双曲型微分方程的一类高精度数值方法, 其在并行计算、处理复杂几何和自适应网格细化 (hp-adaptivity) 等方面具有天然优势. 近年来, 针对动力学方程的渐近保持半拉格朗日格式得到广泛研究, 详见相关文献 [67–70] 等, 和本文不同的是其考虑的极限不包含对流速度的刚性; 关于半拉格朗日间断有限元格式的具体构造, 可参考文献 [71–73]. 记具有 p 阶时间精度和 q 阶空间精度的半拉格朗日间断有限元方法为 SL_pLDGq [45], 并对比基于交错网格的二阶显式球谐函数法 StaRMAP [74]. 本节展示的数值结果均来源于文献 [45].

1) 三维变散射系数问题. 由图2所示, 对于具有剧烈空间变化的散射系数分布, 特征线方法利用沿特征方向追踪信息的机制, 在大时间步长约束的条件下仍能保持稳定可靠的计算结果; 相比之下, 传统显式格式在相同时间步长条件下呈现数值不稳定性, 难以得到有效的数值结果.

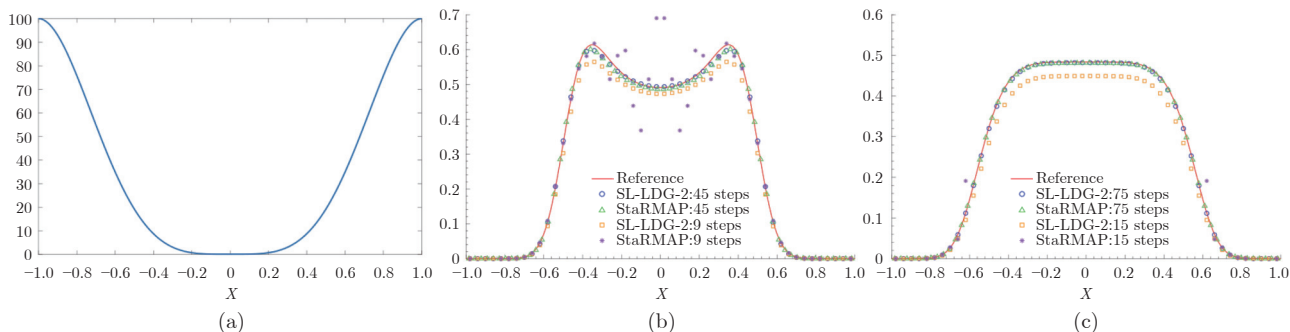


图 2 三维变散射问题. (a) 散射系数分布图; (b)(c) 特征线法与球谐函数法的对比图.

Fig. 2 Three-dimensional varying scattering problem. (a) Scattering distribution; (b)(c) Comparison of characteristic and spherical harmonic solutions.

2) 三维点源问题. 由图3所示, 当分布函数剧烈变化时, 球谐函数法虽然能够保持解的旋转不变性, 但是容易产生非物理数值振荡; 而离散速度法由于严格保持输运结构, 不容易产生波动. 由表1可见, 本文所提方

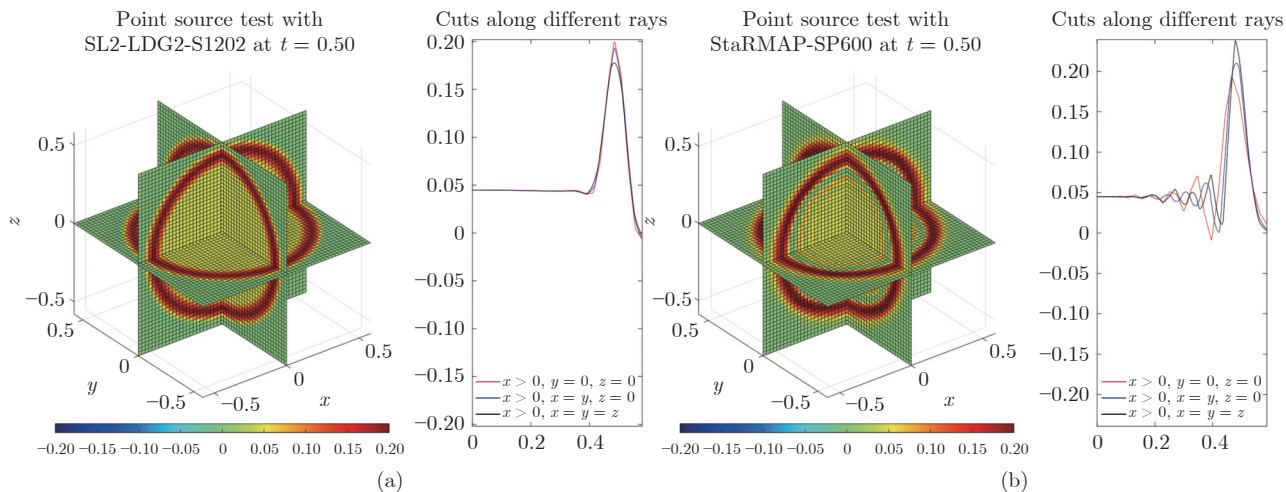


图 3 三维点源问题 ($\sigma = 1$). (a) 球谐函数法; (b) 离散速度法.

Fig. 3 Three-dimensional point source problem ($\sigma = 1$). (a) Spherical harmonic method; (b) Discrete velocity method.

法的并行效率高, 其加速比随计算核心数增加依然接近线性增长, 验证了前文的理论分析结果.

2.2 非线性辐射输运方程

对于非线性辐射输运方程, 基于粒子的蒙特卡罗方法是主流的数值模拟工具之一. 在源项的离散上, 显式蒙特卡罗方法受到极为严格的时间步长限制. 目前, 隐式蒙特卡罗 (IMC, Implicit Monte Carlo) [66] 是应用最为广泛的算法. 该方法将光子的吸收与再发射过程近似为“伪散射”, 虽显著增强了计算的稳定性, 但在光学厚介质区域会引入高昂的计算开销. 为提高计算效率并规避伪散射的引入, 我们将基于特征线近似的宏观模型与传统蒙特卡罗方法相结合, 构造了新型蒙特卡罗方法 [46,47] (EMC), 并对比具有相同参数的 IMC 方法.

1) Infinite medium 问题(方差测试). 本算例求解的是灰体辐射输运方程, 数值结果来源于文献 [47]. 我们模拟一个处于扩散尺度的无限平板问题, 真实解应该保持恒定温度. 定义方差与计算时间的性能指标 (FOM, Figure of Merit) 为

$$FOM = \frac{1}{VAR \times RT} \text{ ,}$$

其中, VAR 是方差, RT 是 CPU 运行时间. 更高的 FOM 代表着更好的综合计算性能. 当模拟时间 $t = 1.0$ ns 时, 如表 2 所示, IMC 方法耗时 669 秒, EMC 方法仅耗时 53 秒. 由图 4 所示, 虽然 EMC 方法的温度变化相较于 IMC 方法更加剧烈, 但是其 FOM 指标仍表现出明显优势, 这表明 EMC 以略高的方差代价换取了计算效率的提升. 在光学厚区域中, IMC 方法引入的伪散射机制会显著延长粒子的有效存活时间, 使辐射能量缓慢沉积到物质场中, 因而温度更新时的统计波动较小. 相较而言, EMC 方法基于宏观方程对发射源项进行估计并直接抽样, 大量的粒子权重快速衰减至 0, 意味着辐射能量更加迅速地沉积到物质场中, 从而使温度更新容易表现出更大的瞬时波动.

表 1 三维点源问题的 OpenMP 并行效率测试.

Table 1 Parallel efficiency test for the three-dimensional point source problem with OpenMP.

线程数	运行时间(秒)	加速比	并行效率
1	13 552.66	—	—
2	6 835.17	1.98	99.14%
4	3 512.93	3.86	96.45%
8	1 641.28	8.26	103.22%
16	1 047.95	12.93	80.83%

表 2 Infinite medium 问题 CPU 运行时间对比.

Table 2 CPU runtime comparison of Infinite medium problem.

算例	EMC(秒)	IMC(秒)
Infinite medium	53	669

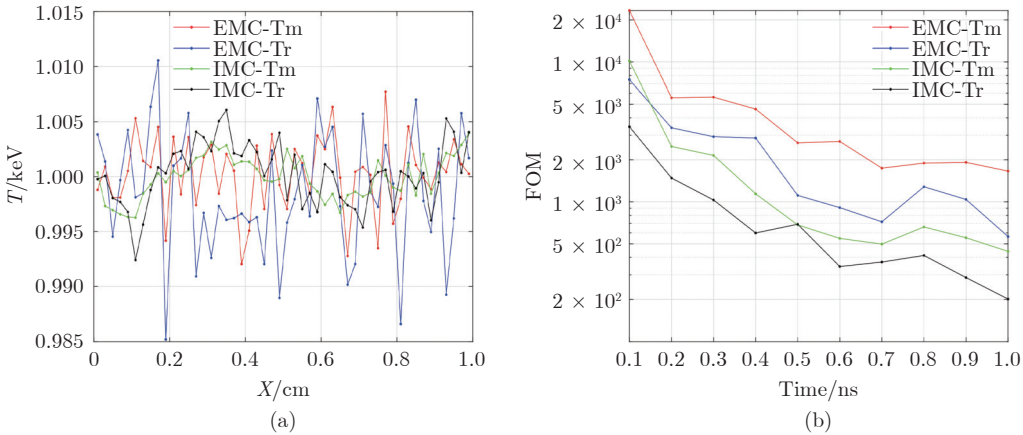


图 4 Infinite medium 问题. (a) 温度对比; (b) FOM 指标对比.

Fig. 4 Infinite medium problem. (a) Comparison of temperature. (b) Comparison of FOM.

2) Marshark-2B 问题. 本算例求解的是灰体辐射输运方程, 数值结果来源于文献 [46]. 由表 3 和图 5 可知, 对于扩散占优的问题, 在参数相同的情况下, EMC 与 IMC 的数值解高度吻合. 尽管 EMC 方法的方差稍大, 但其计算效率具有显著优势.

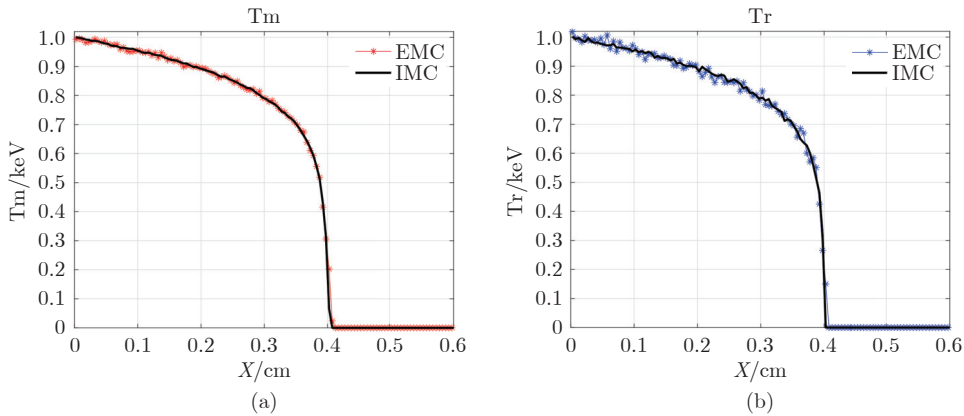


图 5 Marshark-2B问题. (a) 物质温度; (b) 辐射温度.

Fig. 5 Marshark-2B problem. (a) Material temperature; (b) Radiation temperature.

3) 多群空腔问题. 本算例求解的是频率依赖辐射输运方程, 数值结果来源于文献 [47]. 由表 4 和图 6 可知, 对于阻尼系数间断的问题, EMC 与 IMC 在相同参数下的数值解同样高度吻合. 虽然 EMC 的统计波动较大, 但其计算速度显著提升.

表 3 Marshark-2B 问题 CPU 运行时间对比.

Table 3 CPU runtime comparison of Mashark-2B problem.

算例	EMC(秒)	IMC(秒)
Marshark-2B	1 578	14 694

表 4 多群空腔问题 CPU 运行时间对比.

Table 4 CPU runtime comparison of the multi-group

hohlraum problem.		
算例	EMC(秒)	IMC(秒)
Hohlraum	17 687	96 332

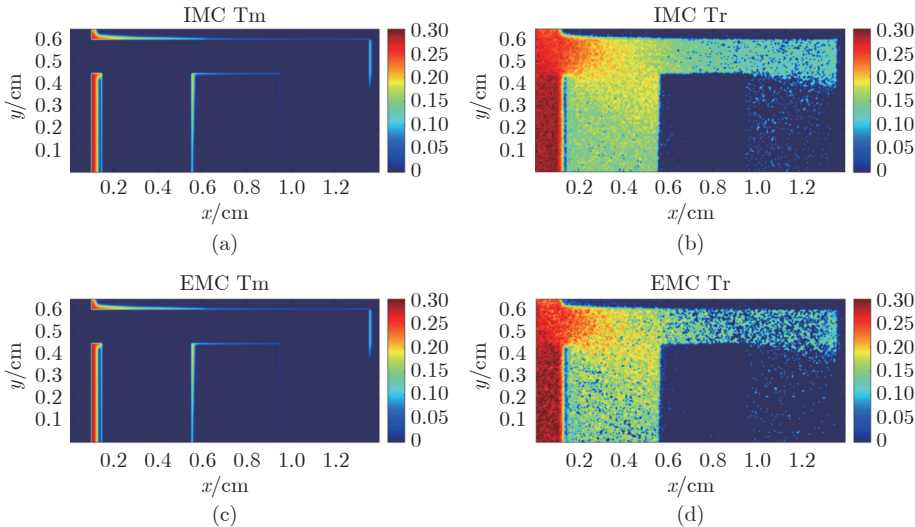


图 6 多群空腔问题. (a) IMC 物质温度; (b) IMC 辐射温度; (c) EMC 物质温度; (d) EMC 辐射温度.

Fig. 6 Multi-group hohlraum problem. (a) IMC material temperature; (b) IMC radiation temperature; (c) EMC material temperature; (d) EMC radiation temperature.

3 结论

本文介绍了一类基于特征线回溯思想构造的渐近保持格式, 并展示其在辐射输运问题中的应用. 该方法的理念是沿特征线构建尺度自适应的宏观-微观耦合模型, 以还原动理学模型的渐近极限. 在数值计算过程中, 先求解宏观方程以获得宏观量的预测值, 再求解微观方程并对宏观量进行相容修正, 形成闭合的迭代框架. 该方法具备全尺度一致的无条件稳定性, 优异的并行效率和良好的扩展性, 适合求解高维跨尺度动理学问题.

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Efficient Asymptotic-Preserving Method Based on Characteristics for Kinetic Equations

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Abstract: Kinetic equations serve as fundamental mathematical models for describing the dynamic behavior of particle systems. Their intrinsic multiscale structure, however, poses significant challenges in developing efficient and robust numerical schemes. A popular approach is to design asymptotic preserving (AP) schemes. It aims to create a unified solver across the computational domain that captures the correct asymptotic limits at a discrete level. In recent years, we have developed a class of unconditionally stable AP schemes based on characteristic backtracking. These methods feature low computational complexity, high parallel efficiency, and excellent scalability, making them particularly suitable for long-time simulations of high-dimensional, multiscale transport phenomena. Owing to these advantages, the proposed framework holds great potential for applications in national defense engineering, astrophysics, semiconductor, and biochemistry.

Key words: kinetic equation; parallel computing; characteristic backtracking; asymptotic preserving